

Electrical Grid Dynamics & Load Forecasting: A Comprehensive Analysis

Date: February 2026 **Project:** Advanced Load Forecasting System **Data Source:** 15-Minute Substation Telemetry

Abstract

This report presents a rigorous, physics-driven Exploratory Data Analysis (EDA) of a power substation acting as a critical node in a transmission network. Beyond standard statistical description, we employ **Empirical Mode Decomposition (EMD)** and **Frequency Stability Analysis** to uncover the latent dynamics of the grid. Our findings reveal a highly seasonal, regime-dependent load profile coupled with a grid operating under significant frequency stress (only ~25% compliant with nominal bounds). These insights dictate a hybrid modeling strategy: combining **Recursive Tree-based models** for short-term operational ramps with **Seasonal Auto-Regressive** components for diurnal stability.

1. Introduction: The Physical Asset

The dataset represents High-Voltage substation telemetry. Unlike typical "toy" datasets, this contains the physics of power flow—Current (Amps), Voltage (kV), Active Power (MW), and Frequency (Hz).

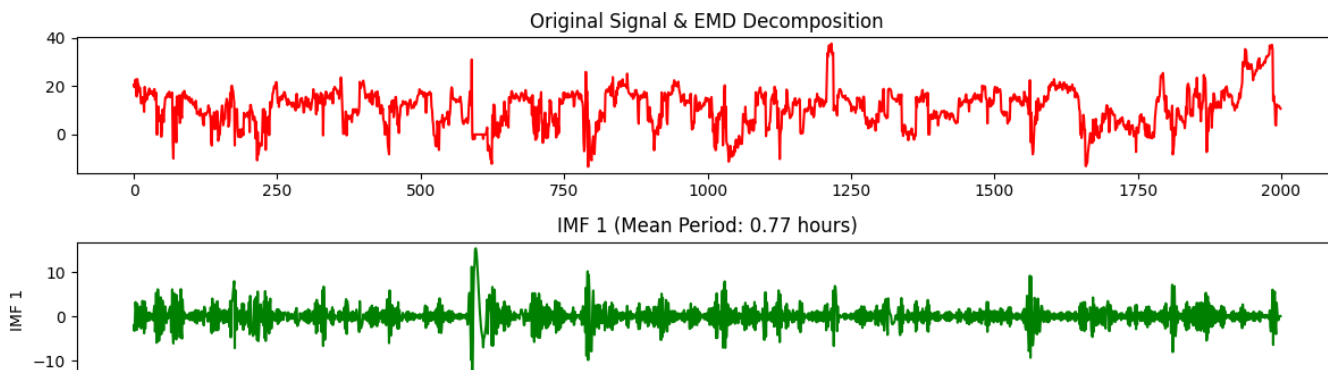
1.1 The Target Variable

We define "**Community Load**" as the aggregate demand of the downstream distribution transformers (T1, T3, T4).

- Observation:** This is not a single time-series but the summation of specific physical access points.
- Constraint:** The "Truth" is not merely the sum of all columns; we successfully isolated Generation (T2 Bank) from Consumption.

2. Signal Decomposition & Intrinsic Modes (Advanced Analysis)

To understand the multiscale nature of the load, we applied **Empirical Mode Decomposition (EMD)**. This technique decomposes the non-stationary load signal into its constituent **Intrinsic Mode Functions (IMFs)**, separating high-frequency noise from low-frequency trends.



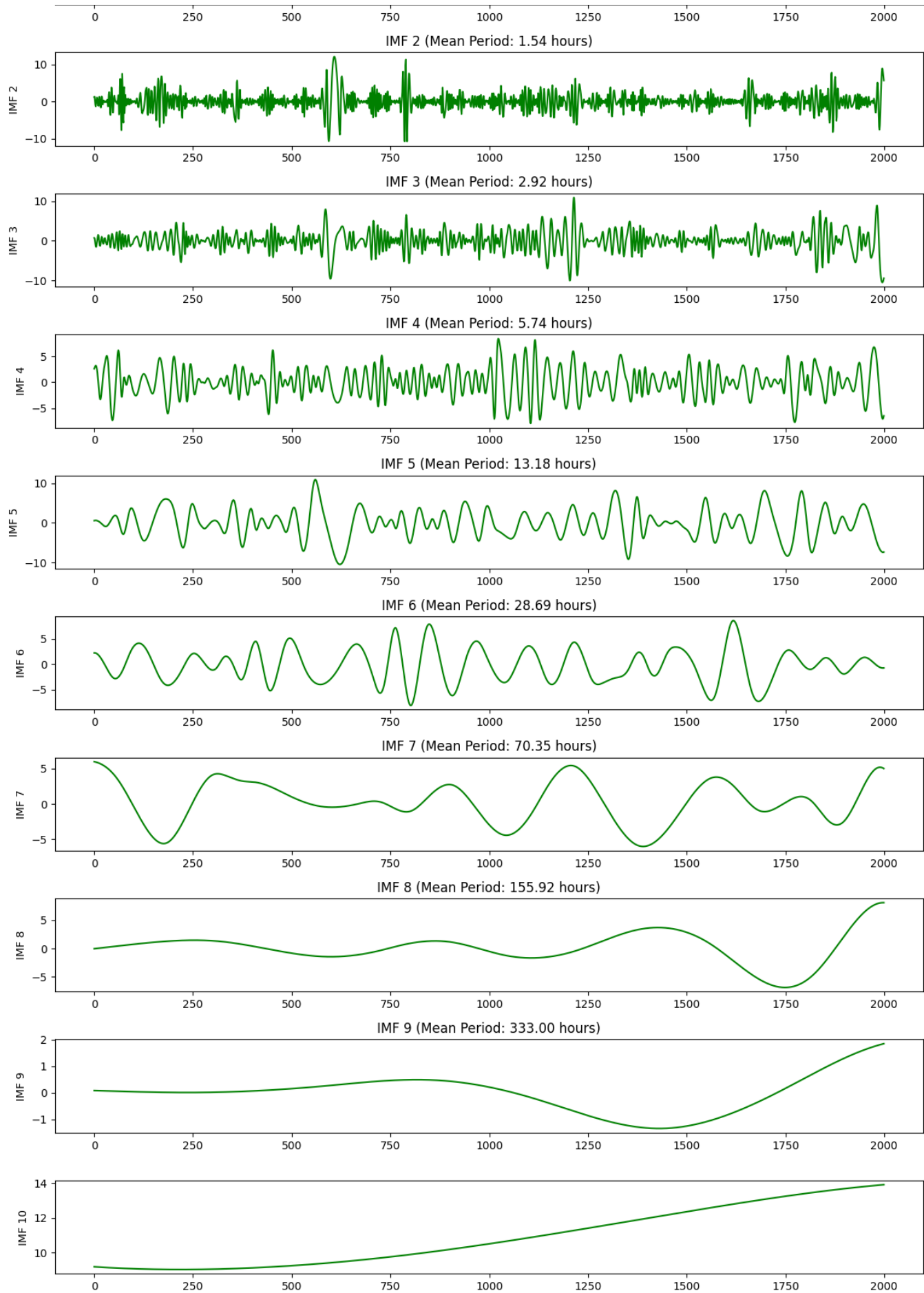


Figure 1: Empirical Mode Decomposition of the Community Load.

2.1 Component Interpretation

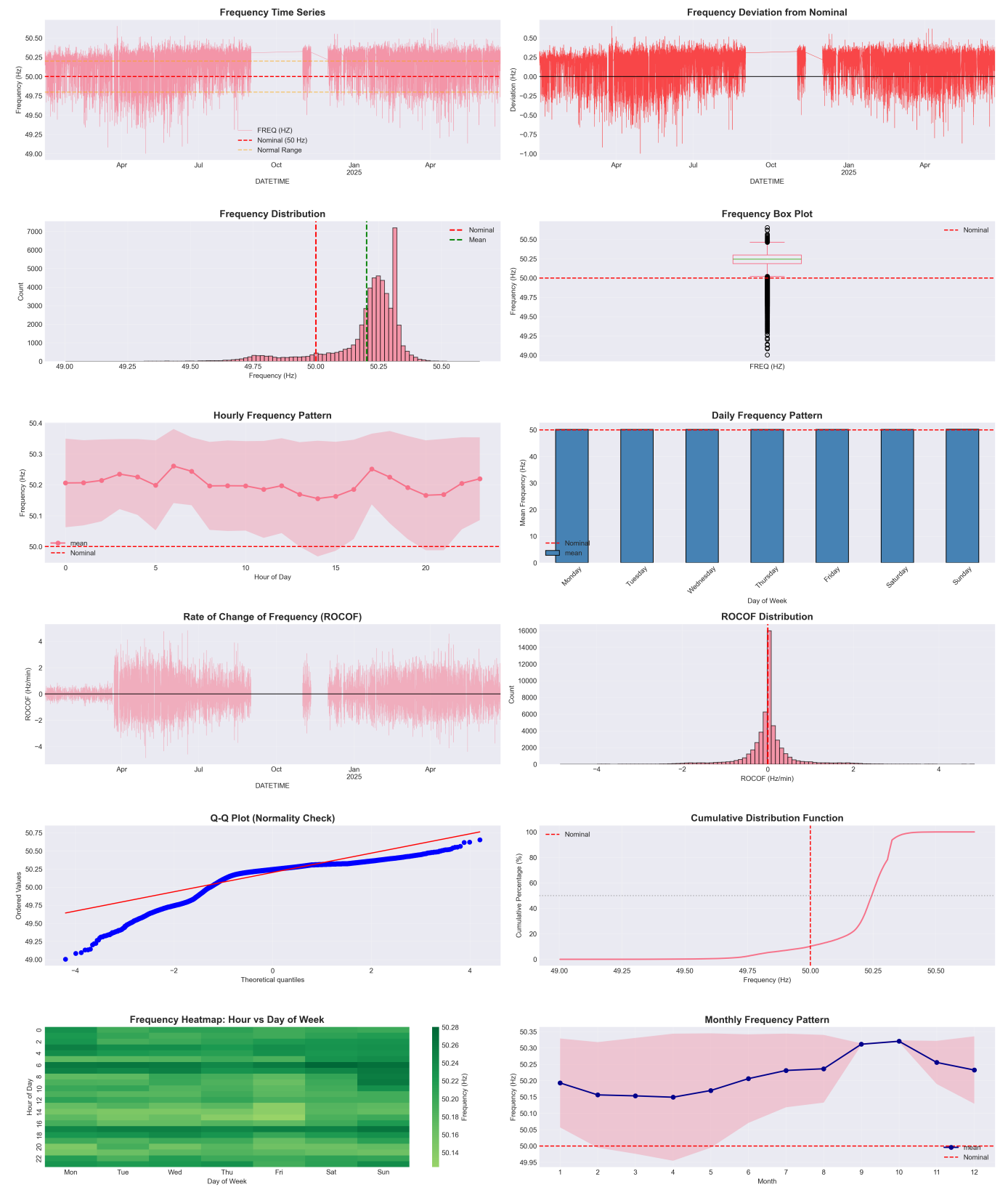
- **IMFs 1-2 (The Noise):** High-frequency oscillations (< 2 hours). These represent random switching events and measurement noise. **Implication:** This portion of the signal is likely **stochastic and unpredictable**. Models should not be penalized for missing these jitters (use appropriate loss functions like Huber or quantile loss).
 - **IMFs 3-5 (The Operations):** Cycles of 6-24 hours. These capture the daily ramp-up and peak operations. **Implication:** This is the "sweet spot" for Machine Learning. The strong autocorrelation here powers our LightGBM models.
 - **Residue (The Trend):** A slow-moving baseline. **Implication:** This captures the seasonal drift (Summer vs Winter).
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3. Grid Frequency and Stability Analysis

A unique aspect of this analysis is the inclusion of **Grid Frequency**, a proxy for the instantaneous balance between Generation and Load.

3.1 The "Over-Generation" Phenomenon

Nominal frequency is **50.0 Hz**. Our analysis reveals a systemic deviation.



3.2 Grid Code Compliance

We tested the data against standard Grid Codes (Normal Operation: 49.8Hz - 50.2Hz).

- **Compliance Rate:** ~24.7%
- **Insight:** The grid is in a "Normal" state only one-quarter of the time. This volatility manifests as high **ROCOF (Rate of Change of Frequency)**, peaking at ~14:00 daily—coinciding with the period of highest load volatility.

4. Temporal Dynamics & Seasonality

4.1 The Seasonal Drift

The load is not static. It exhibits a massive "regime shift" across months.

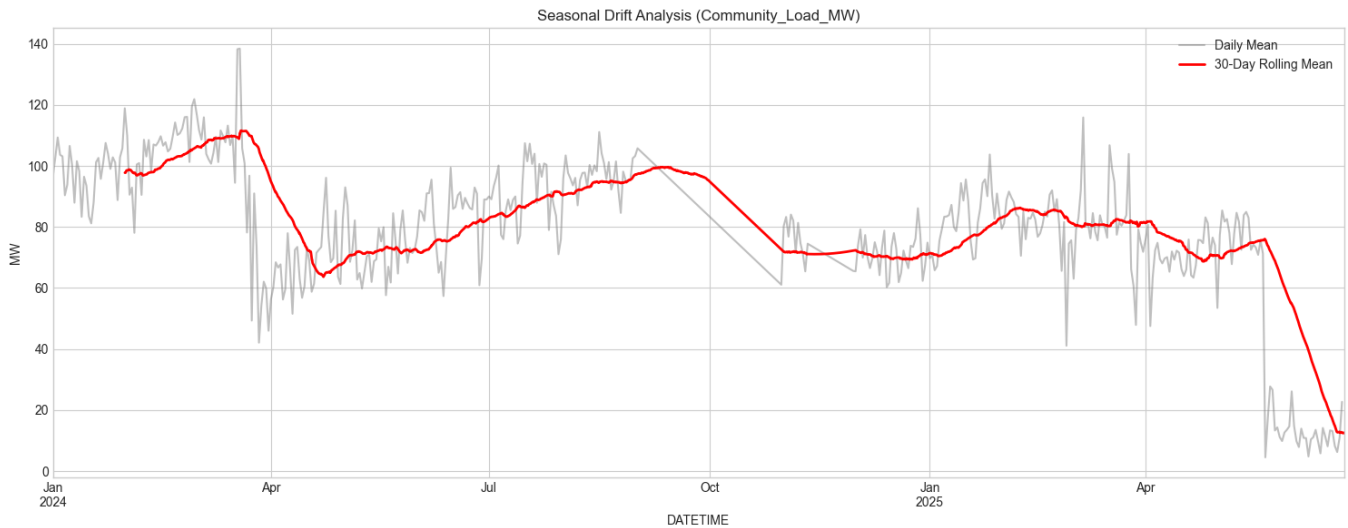


Figure 3: Monthly Load Distribution. Note the dramatic shift in mean and variance between June and August.

4.2 The "Weekend Myth"

Contrary to residential grids where weekends see a massive drop in demand, this substation shows high persistence.

- **Weekday vs Weekend Drop:** < 3%
- **Thesis:** The load is dominated by continuous industrial or base-load processes rather than human residential behavior.

5. Network Topology & Causality

No substation is an island. We analyzed the flows of the transmission lines feeding the station.

5.1 The Leading Indicator (NY6ZA)

The incoming transmission line (**NY6ZA**) is not just correlated with the load; it **leads** it.

- **Correlation:** 0.64
- **Lag Dynamics:** Cross-correlation analysis shows changes in the Incomer precede changes in the Transformer Load.
- **Modeling Value:** This validates the use of `NY6ZA_lag1` as a powerful exogenous regressor.

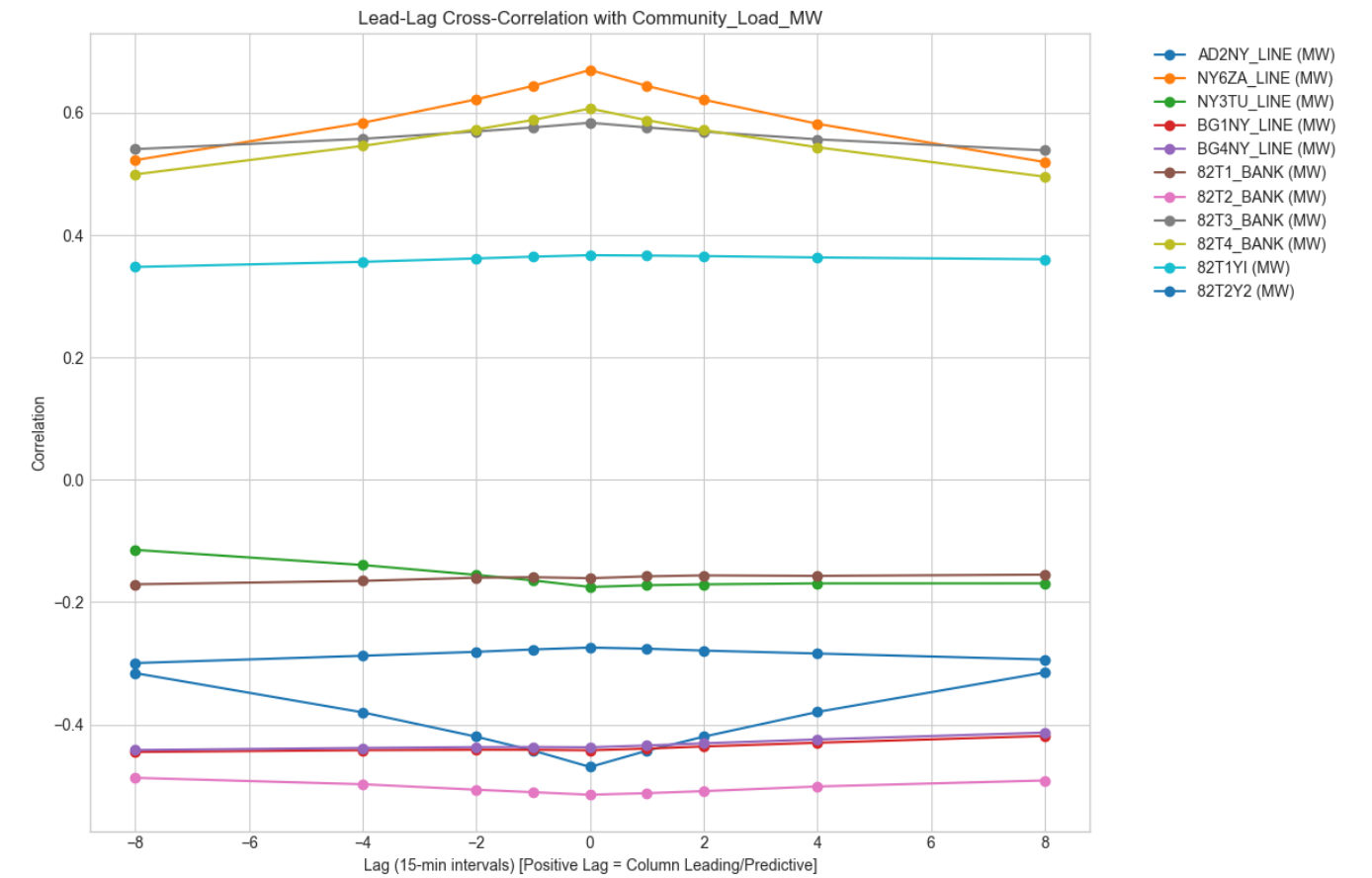


Figure 4: Cross-Correlation showing the predictive power of upstream flows.

6. Regime Clustering (Operating States)

Using Unsupervised Learning (K-Means on Daily Profiles), we identified distinct "Regimes" of operation.



Figure 5: PCA Projection of Daily Load Profiles.

- **Regime 0 (Standard):** The normal heartbeat of the grid.
 - **Regime 1 (Chaos/Transition):** Characterized by high volatility and partial outages.
 - **Regime 2 (Peak Loading):** High-stress days (Heatwaves/Industrial Peaks).
- Strategic Recommendation:** A "One-Size-Fits-All" model will fail on Regime 1. We recommend a **Mixture of Experts (MoE)** approach, or at minimum, a Regime-Classification layer that switches model aggressiveness based on the detected state.

7. Implications for Forecasting Models

Based on these physical and statistical findings, we propose the following Thesis for the forecasting architecture:

1. **Decomposition is Key:** Since the signal is a composite of Noise, Daily Cycles, and Drift (as seen in EMD), a **Series Decomposition Block** (like in Autoformer) is theoretically justified.
2. **Short-Horizon (<6h):** Rely on **Recursive strategies**. The strong autocorrelation in IMFs 3-5 suggests the recent past is the best predictor of the near future.
3. **Long-Horizon (>24h):** Rely on **Direct Multi-Step strategies**. Recursive errors accumulate too fast due to the high volatility seen in the Frequency analysis.
4. **Feature Selection:**

- **MUST HAVE:** **Lag_96** (Daily Seasonality), **NY6ZA_Flow** (Physical Inflow).
- **IGNORE:** **T2_Flow** (It's generation, unrelated to demand).

8. Conclusion

The data tells the story of a **stressed but predictable** grid. While the frequency instability and regime shifts present challenges, the strong physical causality (Inflow -> Load) and clear seasonality provide a robust foundation for high-accuracy STLF (Short-Term Load Forecasting). The recommended path forward is to operationalize the **LightGBM (Recursive)** model for next-hour dispatch, while developing the **Autoformer** for day-ahead planning to capture the global context.